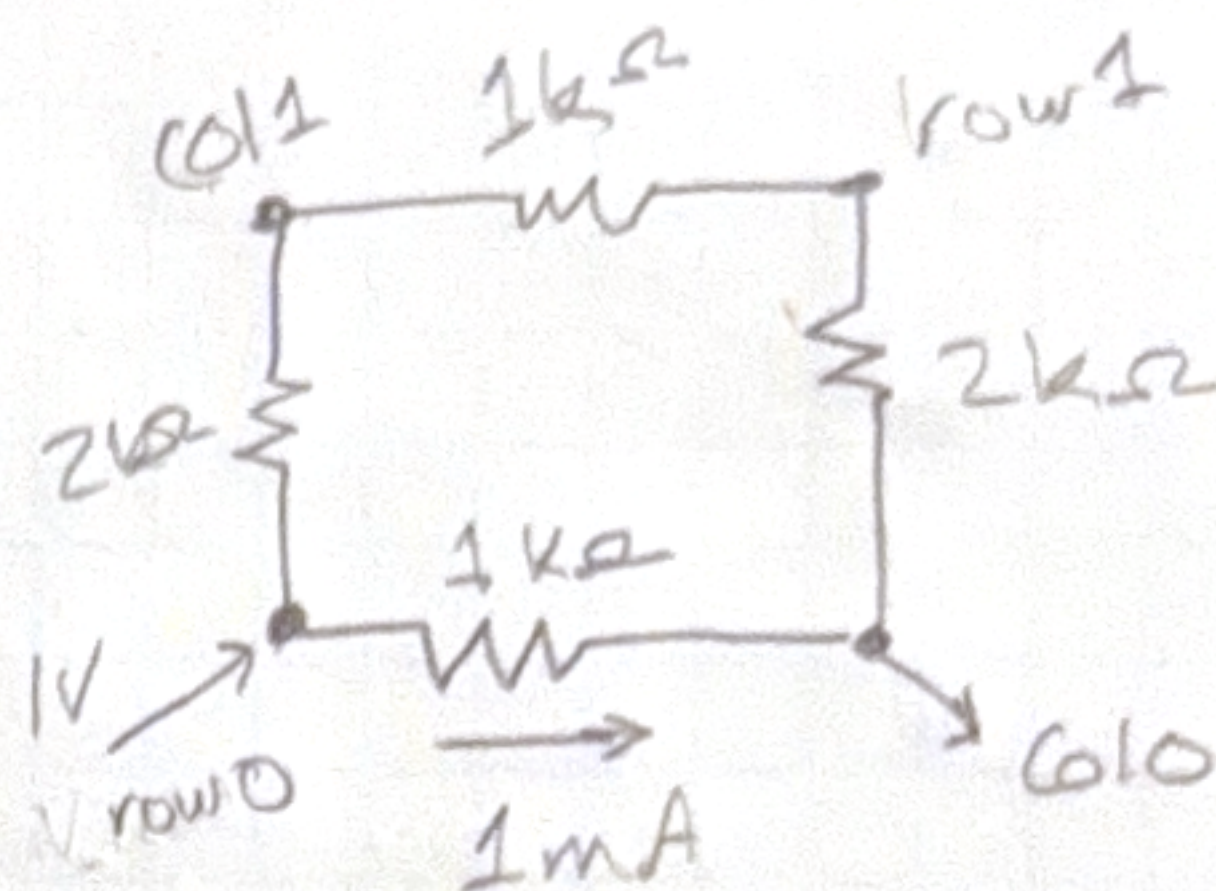
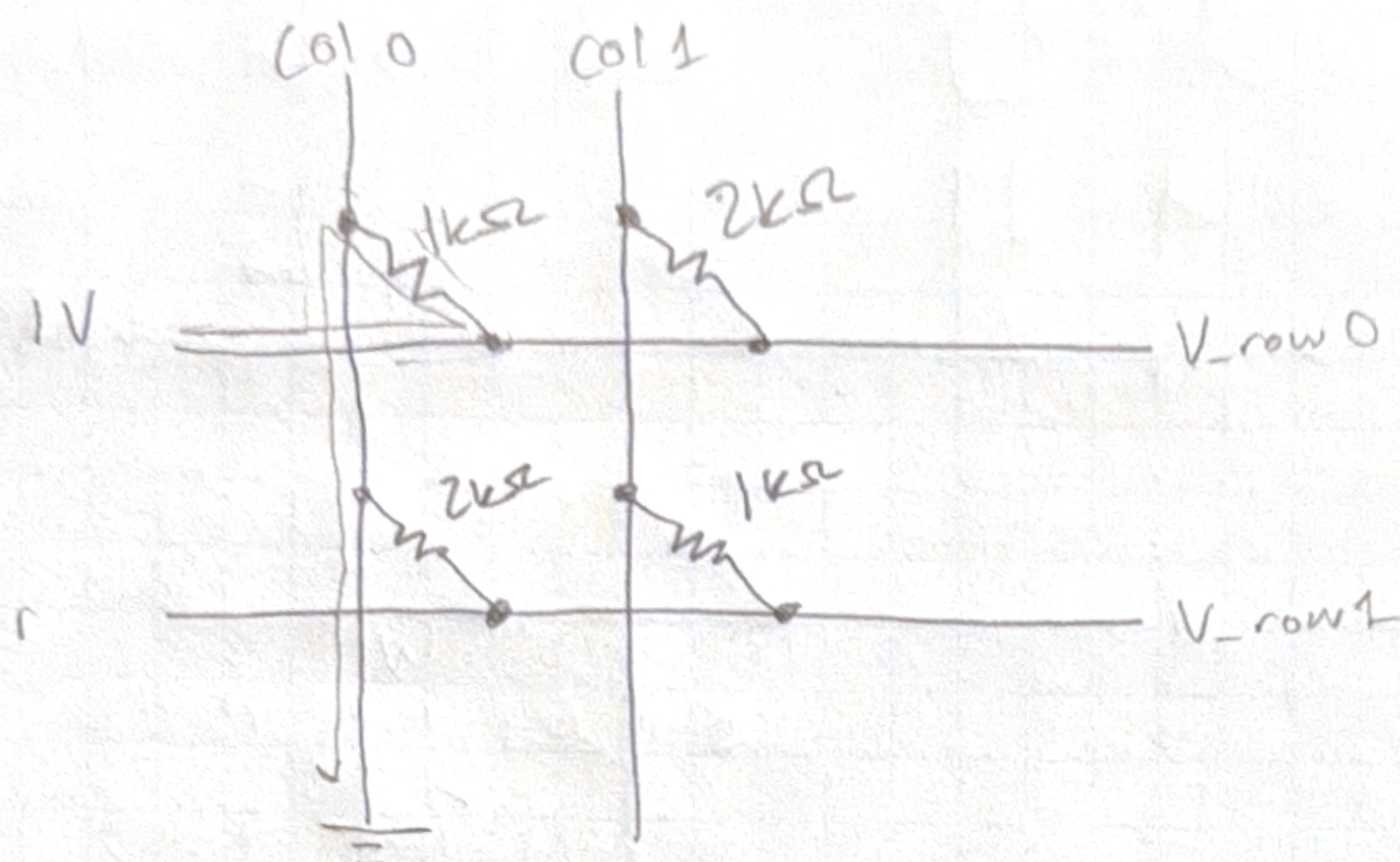




$$1. \quad I_{\text{col}0} = \frac{1V}{1k\Omega} = \boxed{1mA}$$

$$2. \quad I_{\text{col}0} = \frac{1V}{1k\Omega} + \frac{1V}{5k\Omega} = \boxed{1.2mA}$$

$$V_{\text{row}1} = 0.2mA \cdot (2k\Omega + 1k\Omega) = \boxed{0.6V}$$

$$V_{\text{col}1} = 0.2mA \cdot 2k\Omega = \boxed{0.4V}$$

3. Ideally, current flows only through the shortest/least resistive path, but in reality current can flow through any available path. In this case, this "sneak current" is relatively small since there is only one path. In a large array, there will be many additive paths which will affect the expected result.